

3D Computer Vision: HW3

Human Pose Estimation with SMPLify

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1 Pipeline of SMPLify¹

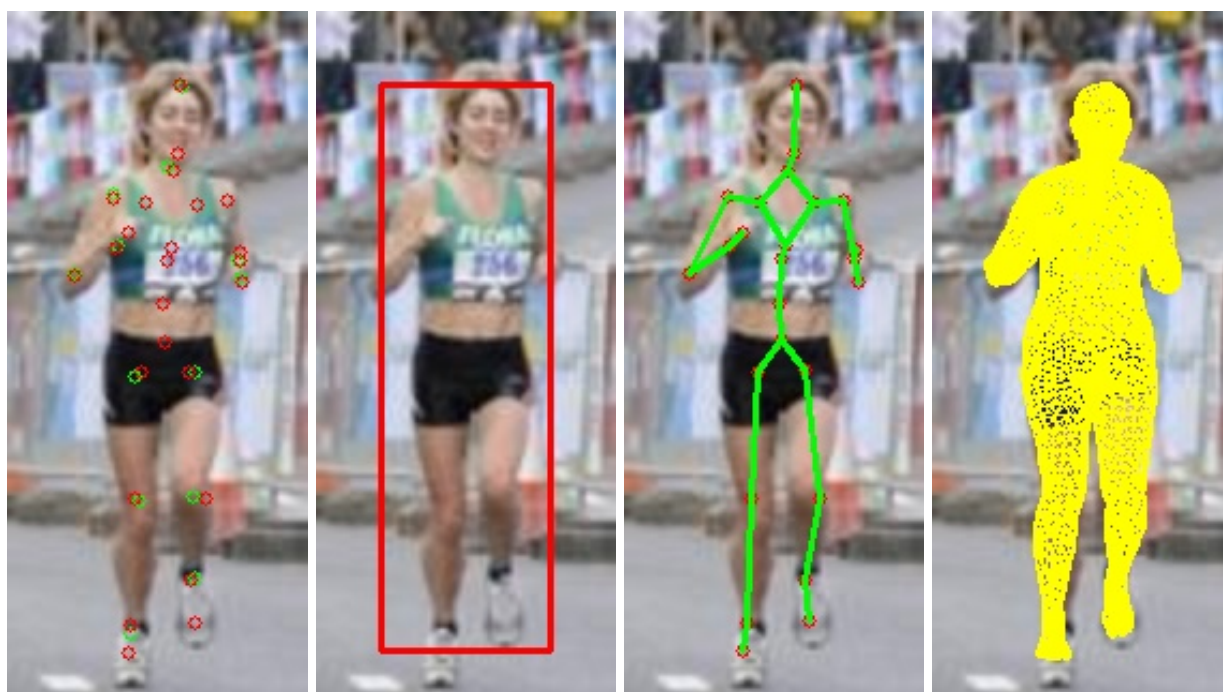


Figure 1: Pipeline for SMPLify images. From left keypoint extraction, bounding box drawing, skeleton drawing and mesh extraction.

SMPLify, a fully automatic pipeline to estimate the 3D pose and shape of the human body is an integrating method for 3D human pose estimation. The paper presents a method for estimating 3D human pose and shape from a single 2D image.

The input to the SMPLify model is a single 2D image of a person. The model assumes that the person is the dominant subject in the image, and assumes that the intrinsic parameter of the camera is known.

The first step in the pipeline is to detect 2D keypoints of the human body in the input image. The authors use a state-of-the-art 2D keypoint detector, such as DeepCut or OpenPose, to extract the 2D locations of body joints. The detected 2D keypoints serve as the initial input for the subsequent steps. These ground-truth keypoints are denoted in Figure 1-1 as green circles (total 13 joints, not including the neck and head top).

The SMPL (Skinned Multi-Person Linear) model is a parametric 3D human body model. It represents the human body as a function of pose parameters (joint angles) and shape parameters (body shape coefficients). The SMPL model is used as the underlying representation for estimating the 3D pose and shape of the person.

The SMPLify model defines an objective function that measures the discrepancy between the projected 3D

¹Bogo et al., Keep it SMPL: Automatic Estimation of 3D Human Pose and Shape from a Single Image, ECCV 2016

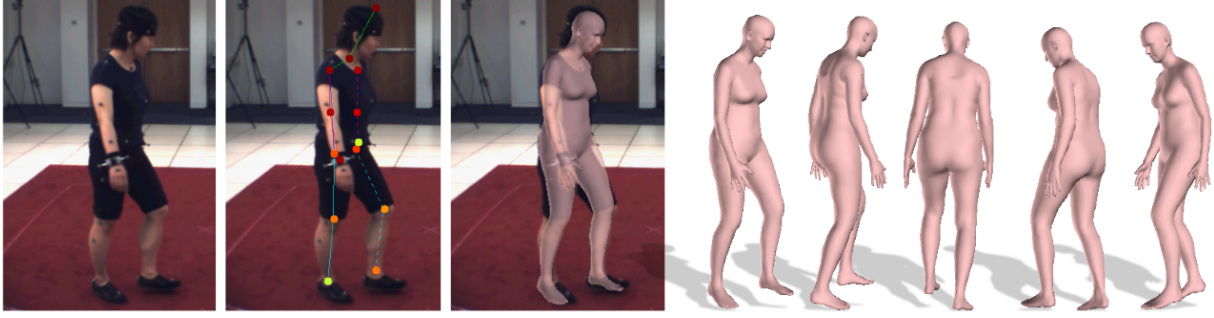


Figure 2: System overview from the SMPLify paper[1]. From left to right: Target image, CNN-based joint prediction, 3D body model matching, estimation of 3D body shape and pose.

keypoints of the SMPL model and the detected 2D keypoints in the image. The objective function consists of several terms:

- Joint-based data term: Measures the difference between the projected 3D keypoints and the detected 2D keypoints.
- Prior term: Encourages realistic poses and shapes by penalizing unlikely or impossible configurations.
- Interpenetration term: Penalizes self-intersections of the body to ensure a plausible 3D reconstruction.

The SMPLify model optimizes the pose and shape parameters of the SMPL model to minimize the objective function. The optimization is performed using a gradient-based approach, and the prior hyperparameters are tuned so that after each iteration, less pose prior gives effect to the training.

The output of the SMPLify model is the estimated 3D pose and shape of the person. The pose is represented by the joint angles of the SMPL model, and the shape is represented by the shape coefficients. The estimated 3D pose and shape can be used to render a 3D mesh of the person, and the projection is shown in Figure 1-4. The whole process is given in Figure 2 from the original paper.

2 Results

2.1 Image results

The output keypoint and bounding box drawings for selected few images are shown in Figure 3. The full data is under `results/bbox_keypoints_{i}.jpg`.

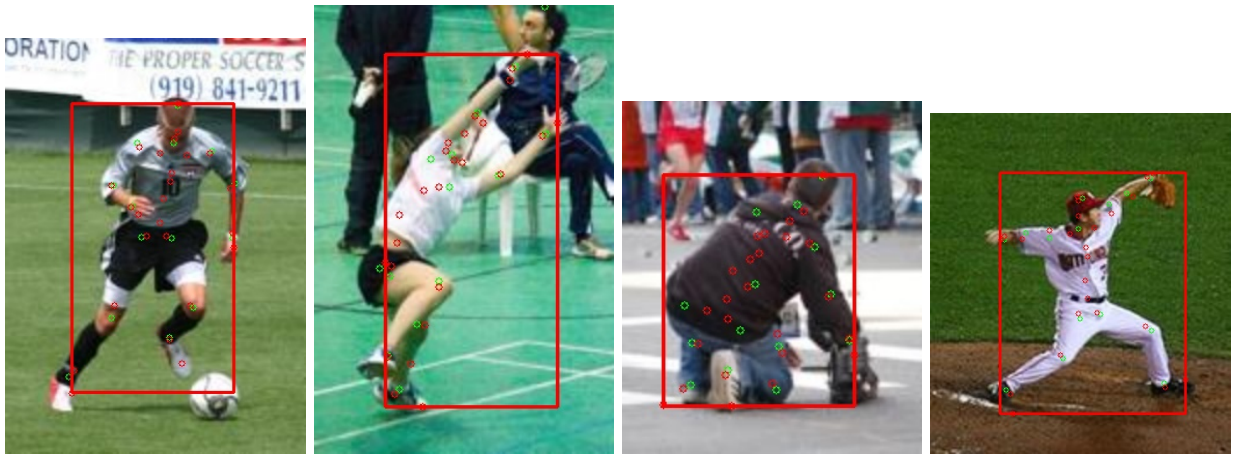


Figure 3: Keypoint and bounding box drawings for images 1, 9, 10, 15

The output skeletons for selected few images are shown in Figure 4. All 25 joint points for SMPLify joints are shown in red. The full data is under `results/skeleton_{i}.jpg`.

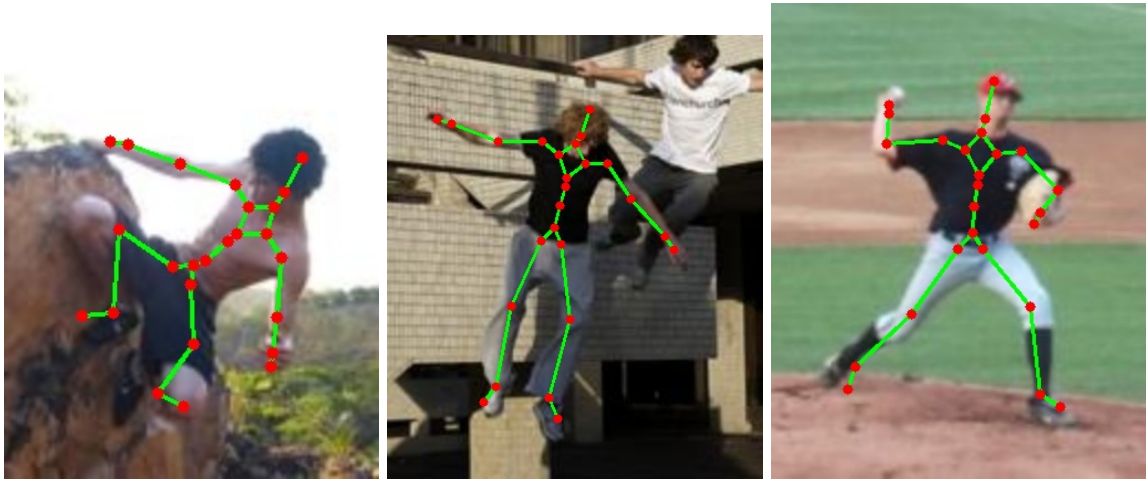


Figure 4: Skeleton for images 2, 7, 19

The projected mesh for selected few images are shown in Figure 5. The full data is under `results/mesh_{i}.jpg` and `results/mesh_{i}.obj`.



Figure 5: Projected mesh for images 4, 5, 13

2.2 Mesh results

The .obj file format is commonly used for storing 3D mesh data, including vertices, faces, and other optional information such as normals and texture coordinates. Each line starting with the letter "v" represents a vertex. The format is "v x y z", where x, y, and z are the coordinates of the vertex. Each line starting with the letter "f" represents a triangular face. The format is "f v1 v2 v3", where v1, v2, and v3 are the indices of the vertices that form the face. In my code `exportMesh` function, this format is used to write to the obj function. I have added 1 to ensure 1-based indexing for obj files.

The following three mesh captures are on Blender, to show triangular mesh shapes better. They are each the first examples in Figure 3, Figure 4 and Figure 5. The full meshes can be found under `results/mesh_{i}.obj`.



Figure 6: Mesh for images 1, 2, 4

3 SMPLify without priors

3.1 Priors in SMPLify

Priors play an important role in 3D reconstruction. Estimating 3D human pose and shape from a single 2D image is an ill-posed problem due to depth ambiguity and occlusions, for example multiple 3D body configurations can project to similar 2D keypoints, leading to ambiguity in the 3D reconstruction. Priors are introduced to constrain the solution space and guide the optimization towards realistic and plausible 3D body configurations. There are a total of 3 priors used in this paper, which are pose priors, shape prior and a penetration term.

The pose prior $E_a(\theta) = \sum_i \exp(\theta_i)$ is the term for unnatural joint angles such as the knee or arms. Also because making a shape with higher probability will result in better results, so a mixture of gaussians about the pose prior, $E_\theta(\theta)$ is used.

The shape prior $E_\beta(\beta)$ captures the statistical distribution of plausible body shapes. During optimization, the shape prior term in the objective function penalizes extreme or unrealistic shape parameters.

The penetration term minimizes the error of 3D vertices and cylinder-estimated surface of the human skin. They add losses to poses and shapes that have penetrating limbs that are impossible in real life.

3.2 Reconstruction with excluded priors

I now demonstrate exclusion of each priors, by changing the `opt_weights` term in the `fit3d-optimizeOnJoints` function. The experiments are done for 6 images, where three of them(1, 9, 19) have qualitatively good quality and the other three(2, 5, 7) have relatively bad quality, to observe differences.

The reconstruction errors with shape, pose and interpenetration priors turned off is as follows for images 1,2,5,7,9,19.

From the table, we can observe the impact of different priors on the reconstruction errors. The observation points are as follows:

- Removing the pose prior (No Pose Prior) tends to increase the reconstruction errors compared to the baselines.
- Removing the shape prior (No Shape Prior) leads to significantly higher reconstruction errors, especially for image 7.
- Removing the penetration error term (No Penetration Error) has minimal impact on the reconstruction errors compared to the Baseline.
- Removing all priors (No Prior At All) results in lower reconstruction errors for most images, but this does not necessarily indicate better 3D body estimation as the priors help ensure plausible and realistic body configurations.

Image Number	Baseline	No Pose Prior	No Shape Prior	No Penetration	No Prior at All
1	4.6853	5.9923	27.4907	4.6817	1.4829
2	13.1575	11.1127	27.9223	13.1575	1.6717
5	16.2983	12.5878	64.1369	16.2983	4.5081
7	12.8007	15.1732	807.2395	12.8007	9.9198
9	13.1574	12.6450	21.0979	13.1574	9.1108
19	7.5091	6.8602	26.7493	7.5091	2.0778
Importance		2	1	3	

Figure 7: Reconstruction values with different priors.

The shape prior in SMPLify is learned from a dataset of 3D human body shapes and captures the statistical distribution of plausible body shapes. Without the shape prior, the optimization process may converge to unrealistic or extreme body shapes that do not resemble human anatomy. The shape prior acts as a regularizer, guiding the estimation towards more reasonable and human-like body shapes. The high reconstruction errors without the shape prior suggest that the 2D keypoints alone are not sufficient to accurately estimate the 3D body shape. The shape prior provides additional information and constraints that are essential for achieving plausible and accurate 3D body estimates.

The penetration error prior is designed to penalize self-intersections of the body parts and prevent physically impossible configurations. However, the results suggest that the other priors (pose and shape) and the data term (joint-based error) are already effective in guiding the optimization towards plausible body configurations, even without explicitly penalizing self-intersections.

By quantitative analysis of the reconstruction errors based on joint projections, we get the importance to be in the order of shape priors, pose priors and finally penetration terms.

However, because the reconstruction errors only consider projected joints, the reconstruction error values are not a perfect score to evaluate the model’s correctness. To observe this, we look at some example result images. The full image set is under `discussion_results`, `no_shape_prior`, `no_pose_prior`, `no_penetration_prior` and `no_all_prior`.

For no pose priors, we can observe that the results have unnatural limb angles and body poses, whereas the overall shape is that of a human. For image 5 and 19, the mesh contrast between baseline results and no pose prior results are shown.



Figure 8: Mesh comparison for images 5, 19

For no shape priors, the reconstruction errors were very big and the reconstructed meshes did not have human form. To understand these effects visually, we compare the skeleton images of the two visualizations.

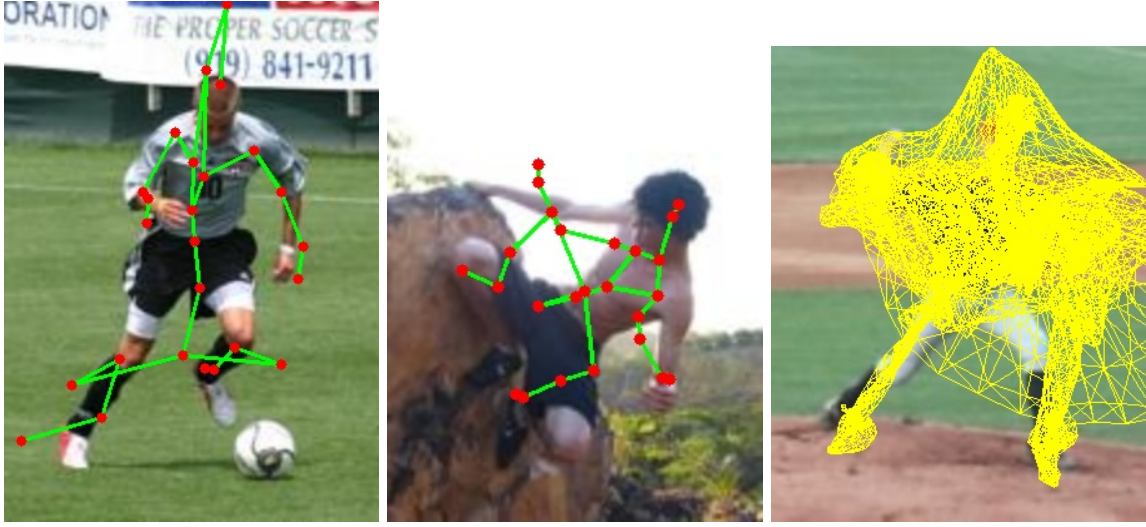


Figure 9: Deformed skeletons for images 1, 2 and deformed mesh for images 19 without shape prior

Finally without penetration terms, the overall reconstruction terms and visual correctness was the best of the three, but there were still ambiguities concerning limb penetrations, such as arms penetrating bodies and therefore having unnatural limb angles. The visualization is best when seen in Blender, so I used blender in Figure 10 to explain this deformation. However in this case, most of the formations were qualitatively correct to the eye, and the speed was fast for inference. We could observe that for bodies with orientations that are parallel to the camera, such that the torso is seen from the side, has lower reconstruction errors and has ambiguity from not using penetration terms, like image 5 as shown below.

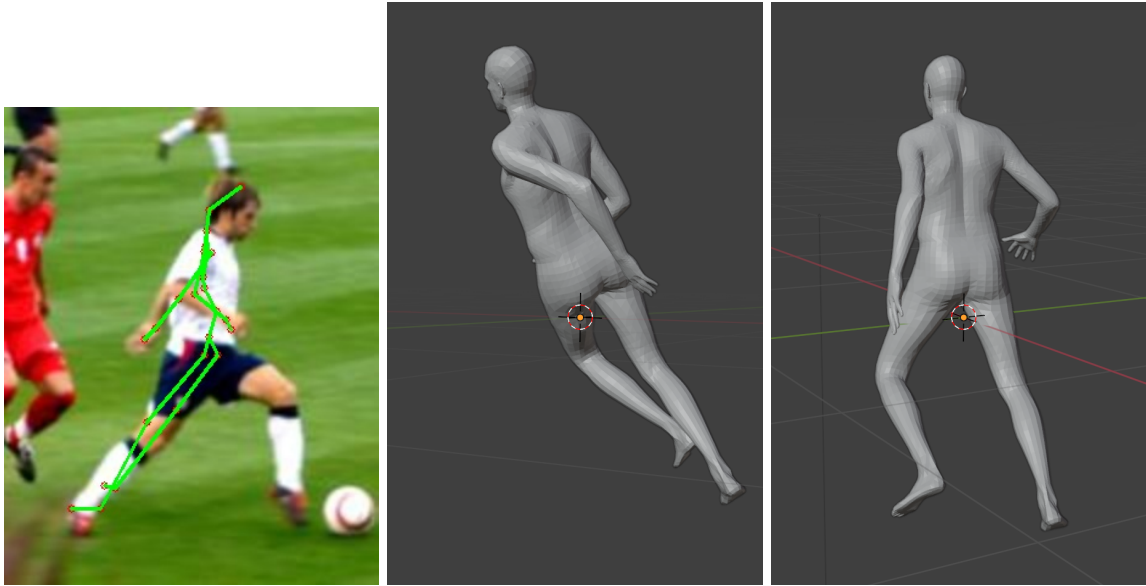


Figure 10: Original image and penetrated arm-body for image 5

Finally the case where no prior is used is demonstrated below, and we can observe that even though with a small reconstruction error, we still get unnatural body shapes, as demonstrated in Figure 11.



Figure 11: Deformed body mesh of image 9 and deformed skeleton of image 5